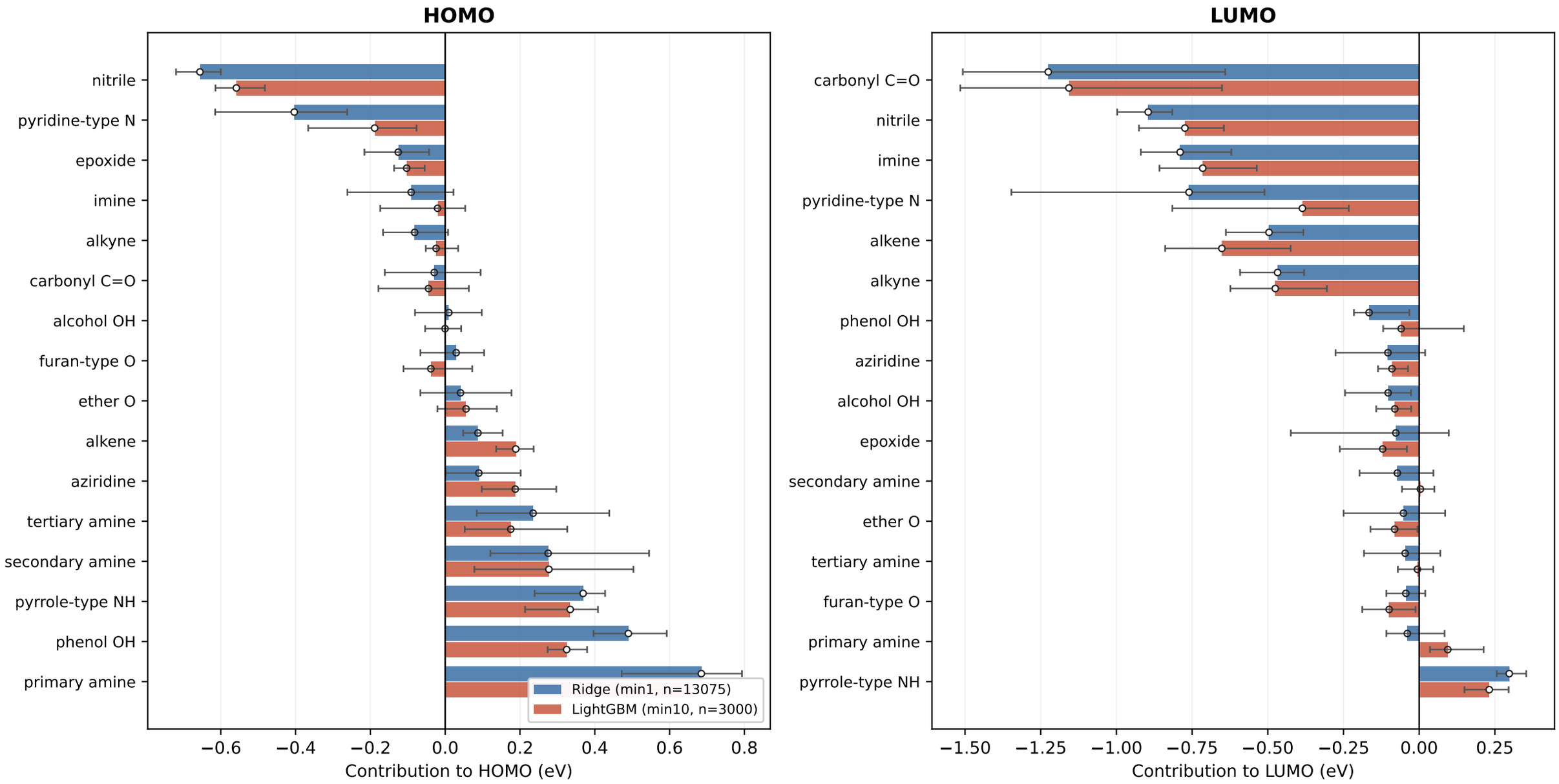
Functional group contributions to frontier orbital energies
Collision-free Morgan fingerprints (r=2), QM9 random-split test set



Bars: median per-molecule SHAP contribution over molecules containing the group. Whiskers: interquartile range (spread = context dependence, not uncertainty).
Filled marker: effect has the same sign in ≥95% of molecules. Carbonyl subtypes merged; see Table X for the LUMO subtype breakdown.